Spatial analysis of frost risk to determine viticulture suitability in Tasmania, Australia

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Abstract

Background and Aims: New sites for viticulture should be located in areas where damaging spring frosts are less frequent and/or severe in order to remain economically viable. The aim of this study was to spatially determine frost risk at a high resolution (80 m) and develop new rules catered to viticultural suitability with respect to frost sensitivity for vines after budburst in spring across the state of Tasmania, Australia.

Methods and Results: Frost risk was mapped for minimum temperature thresholds at −3, −2, −1, 0, 1 and 2°C using temperature values from 636 short-term recording sites and linked to 57 Australian Bureau of Meteorology climate stations. Frost risk was spatially determined using regression tree interpolation and assessed against historical winegrape frost damage records garnered from survey information. Analysis indicated that the frost risk surfaces were accurate in portraying risk at the mesoclimate level and aligned well with grower expectations for frost risk modelled at ≤−1°C.

Conclusions: Classifications of suitable, moderately suitable and unsuitable corresponding to <1 frost every 10 years (<10%), >1/10 to 1 frost every 2 years (10–50%) and 1 > frost every 2 years (>50%) for temperature values ≤−1°C was found to correlate well with viticulture suitability with regard to frost risk (after budburst) in Tasmania.

Significance of the Study: This study presents a methodology for producing high-resolution frost risk maps across large land areas that can accurately identify sites prone to damaging spring frosts and inform on new potential viticulture sites with suitability in mind.

Keywords: *climate, frost, suitability, viticulture, wine*

Introduction

Damage caused by frosts is a significant factor that can limit economic returns on many established vineyard sites. Frosts during spring can injure grapevine shoots or buds leading to low fruiting potential or hinder fruit composition at harvest (Molitor et al. 2014, Schultze et al. 2014). Of particular concern to growers are frosts that occur at or after budburst where actively growing shoots are sensitive to freezing temperature often resulting in severe injury or complete loss of the shoot (Trought et al. 1999, Fennell 2004, Arias et al. 2010). Susceptibility to frost damage also increases as shoots continue to develop thereby enhancing vulnerability (Trought et al. 1999). In temperate or highly continental climates where high levels of air mass variability are encountered, frost damage during this critical stage can be persistent despite having sufficient heat accumulation (Hall and Jones 2010, Yau et al. 2014). From an economic perspective, frost damage has been known to reduce average yield by up to 39% (Molitor and Junk 2011). It is, therefore, important to locate new vineyards in sites where the likelihood of spring frosts are low to prevent future crop loss (Trought et al. 1999).

In Tasmania, Australia, damaging spring frost events have historically been a constraint to winegrape production for existing vineyards sites (Wilson 2001, Halley et al. 2003, Jones et al. 2010). Recently there has been a concerted effort by government to significantly increase Tasmania's agricultural production to A$10 billion by 2050 (Kidd et al. 2015, Tasmania State Government 2015), with the aim to expand the wine industry and identify potential new sites for viticulture (Tasmania State Government 2014). A key requirement is to undertake broadscale suitability analysis and identify potential sites that are inherently less prone to damaging frosts. The current arrangement of working Tasmanian vineyards number around 230 and covers more than 1880 ha (Wine Tasmania 2016). The majority of Tasmanian vineyard plantings is grown in the following regions: Tamar Valley (550 ha), East Coast (322 ha), North East (322 ha) and Coal River Valley (271 ha), which contribute 33, 19, 19 and 16% of state production in 2015, respectively (Wine Tasmania 2015). Smaller regions that include the Derwent Valley (115 ha), North West (26 ha) and Huon/Channel (83 ha) areas make up the remaining 13%. The Pinot Noir cultivar has the highest area of plantings (660 ha) followed by Chardonnay (355 ha) and Sauvignon Blanc (165 ha) (Wine Tasmania 2015). The availability of Tasmanian winegrapes according to the 2015 vintage survey was 10 015 tonnes (Wine Tasmania 2015) with an average retail value of A$14.49/L of wine (Wine Tasmania 2016). The wine sector is an important and growing contributor to the Tasmanian economy that is rapidly expanding with vineyard plantings increasing by more than

doi: 10.1111/ajgw.12314

25% (equivalent to 500 ha) over the past 5 years (Wine Tasmania 2016).

Limited research has been done in successfully delineating frost hazard or frost risk to a high level of detail, despite its importance as a climate variable to winegrape production. Studies by Kurtural et al. (2007) and Schultze et al. (2014) mapped extreme cold temperature values and the number of frost days, respectively, but relied on sparsely located weather station data to produce modelled spatial outputs. Kurtural et al. (2007), for example, used 83 climate stations to guide interpolation at 100 m resolution across the state of Illinois, USA, to produce macroscale climate layers that contributed to a cartographic model to delineate new potential vineyard sites. A shortfall of this analysis was the lack of climate stations to sufficiently map frost at a meaningful level. Frost mapping is best done at the mesoclimate scale due to the strong influence of topographic affects. And such details are often not captured in broadscale analysis (Chung et al. 2006, Yau et al. 2014). A high resolution study by Madelin and Beltrando (2005) demonstrated how it was possible to accurately interpolate minimum temperature values for nights of clear skies with cool wind conditions to produce a frost hazard map for Champagne vineyards with a spatial resolution of 50 m. The mapping used topographic parameters, such as elevation, slope and aspect, to create regression models linked to temperature. The study mapped minimum temperature values over a short period of time—that is relied on five spring seasons to ascertain the hazard map—to identify locations to install frost protection systems for existing vineyards. Although a useful tool for existing winegrowers in the region, the study would benefit from the use of longer-term data to achieve a better indication of long-term viability, particularly from a suitability viewpoint (Hall and Jones 2010).

Webb et al. (2016) showed that using a dense network of short-term temperature sensors could accurately model frost risk for winegrapes at high resolution (30 m) over a longer period of 20 years. This involved deploying 174 short-term recording temperature sensors to a study area measuring 60 000 ha in Tasmania, Australia. A combination of regression kriging and linear least-squares regression enabled long-term climate data to be inferred onto short-term recording sites to produce a high density of calibrated climatic temperature sites. A suite of terrain predictor variables combined with a data mining model allowed frost risk, that is the risk (%) of encountering a frost event ≤–2°C in early spring, to be spatially quantified and validated to a high level of accuracy. The study relied on local industry consultation to formulate the rules for frost risk catered to different degrees of winegrape production suitability. Although it was proven that frost risk could be mapped accurately, the parameters were not empirically tested in regard to the minimum temperature that would predispose grapevines to be most susceptible to frost damage. Hence, there is a need to properly and empirically gauge frost risk catered to the level of perceived crop loss and risk to winegrowers. Kurtural et al. (2007) advocated that such information could be obtained from conducting surveys of existing commercial plantings to gauge the frequency of damaging frost events linked to viticulture suitability.

The aim of this study was to spatially delineate frost risk at a high spatial resolution (80 m) across Tasmania (68 401 km^2) and develop new rules catered to viticulture suitability with respect to frost sensitivity for vines during the critical post-budburst stage in spring. The current study focusses on October which marks the beginning of the Tasmanian growing season, typically running through to April (Hall and Jones 2010). For Pinot Noir—the dominant cultivar for vineyards across the state—budburst typically occurs in late September/early October (Smart and Wells 2014). As measured on the modified Eichorn Lorenz grapevine phenology scale, October generally represents EL03 to EL15 in the grapevine growth stages (Coombe 1995, Jones et al. 2010). The study adopts a multivariate spatial modelling approach prescribed by Webb et al. (2016) to apply frost risk predictions in October across the state of Tasmania by interpolating from 636 short-term temperature logger records, dispersed across the state from 2012, in addition to 57 long-term Australian Bureau of Meteorology (BoM) climate stations. Frost risk was mapped according to minimum temperature thresholds set at −3, −2, −1, 0, 1 and 2°C and aligned to historical records associated with winegrape frost damage using qualitative information garnered from winegrower surveys. From this, we determine a threshold that agrees with growers' knowledge and experience and recommend a new set of frost risk rules with respect to the frequency of damaging frosts (after budburst) linked to viticulture suitability in Tasmania.

Materials and methods

Study area

Tasmania is located off the south-eastern coast of Australia (42.08°S, 146.59°E) covering approximately 68 401 km^2 (Figure 1). The climate is described as cool-temperate with rainfall decreasing gradually from west (>1800 mm/year) to East (<450 mm/year) (Bureau of Meteorology 2015). Frost events occur anytime throughout the year and are most common in the late autumn (May), winter (June to August) and early spring (September) (Table 1). Annually, frost days below 0°C vary from 140 days in the central highland areas with coastal fringes typically having <5 days/year (Figure 1). Within the recognised wine regions, mean annual number of frost days (<0°C) of 6, 8, 9, 9, 10, 11,

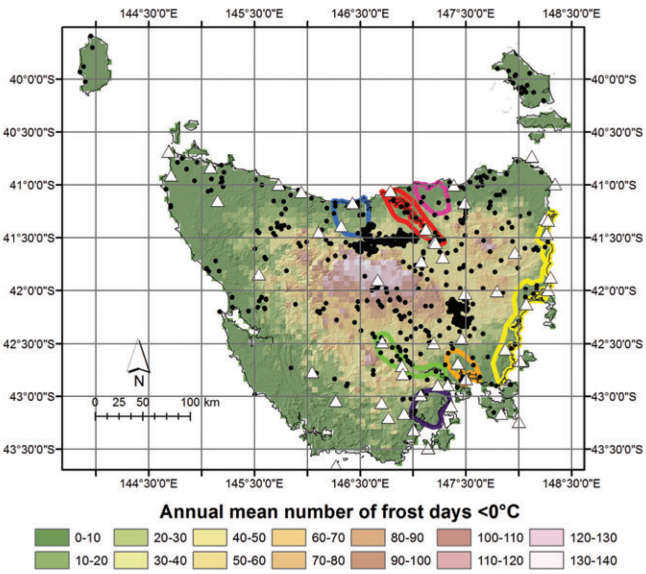

Figure 1. Map of Tasmania showing the annual mean number of frost days below 0°C (Bureau of Meteorology 2017) and the locations of short-term (●) and long-term (△) temperature recording sites as well the locations of recognised Tasmanian winegrowing regions (depicted as coloured polygon shapes): North West (——),Tamar Valley (——), North East (——), East Coast (——), Coal River Valley (——), Derwent Valley (——), Huon/Channel (——).

Table 1. Monthly means for daily minimum and maximum temperature values including the mean number of frost days across Tasmania's winegrowing regions.

		Temperature (°C)						
		East Coast	North East	North West	Coal River Valley	Derwent Valley	Huon/ Channel	Tamar Valley
$\mathrm{Jan}_{(\mathrm{sum})}$	t_{min}	11.0	11.4	11.0	10.8	9.3	9.6	11.4
	t_{max}	21.3	22.0	21.4	22.0	22.5	20.4	22.4
	n_{fd}	0.0	0.0	0.0	0.0	0.0	0.0	0.0
$\mathrm{Feb}_{(\mathrm{sum})}$	t_{min}	11.2	11.7	11.2	10.8	9.4	9.9	11.7
	t_{max}	21.5	22.6	21.9	21.9	22.8	20.6	23.0
	n_{fd}	0.0	0.0	0.0	0.0	0.0	0.0	0.0
$\mathrm{Mar}_{(\mathrm{aut})}$	t_{min}	10.2	10.8	10.0	9.9	8.1	8.9	10.8
	t_{max}	20.0	20.8	20.2	20.3	20.5	18.8	21.1
	n_{fd}	0.0	0.0	0.0	0.0	0.0	0.0	0.0
$\mathrm{Apr}_{(\mathrm{aut})}$	t_{min}	8.0	8.7	7.6	8.0	6.0	7.2	8.6
	t_{max}	17.8	17.8	17.3	17.6	17.2	16.3	18.0
	n_{fd}	0.0	0.0	0.0	0.0	0.0	0.0	0.0
$\mathrm{May}_{(\mathrm{aut})}$	t_{min}	6.0	6.6	5.6	5.6	3.9	5.1	6.4
	t_{max}	16.3	15.8	15.5	15.1	14.8	14.5	15.6
	n_{fd}	1.0	0.0	1.0	0.0	2.0	0.0	1.0
$\mathrm{Jun}_{(\mathrm{win})}$	t_{min}	4.2	4.6	3.5	3.5	1.8	3.3	4.3
	t_{max}	12.8	12.4	12.2	12.0	11.0	11.0	12.4
	n_{fd}	2.0	1.0	3.0	2.0	7.0	3.0	3.0
$\mathrm{Jul}_{(\mathrm{win})}$	t_{min}	3.5	3.9	2.8	3.0	1.2	2.7	3.7
	t_{max}	12.1	11.6	11.5	11.6	10.6	10.4	11.8
	n_{fd}	3.0	2.0	4.0	3.0	9.0	4.0	4.0
$\mathrm{Aug}_{(\mathrm{win})}$	t_{min}	4.1	4.5	3.9	3.8	2.0	3.4	4.5
	t_{max}	13.0	12.5	12.3	12.8	12.2	11.7	12.7
	n_{fd}	2.0	1.0	2.0	2.0	6.0	2.0	2.0
$\mathrm{Sep}_{(\mathrm{spr})}$	t_{min}	5.3	5.6	4.9	5.2	3.6	4.7	5.7
	t_{max}	14.7	13.0	13.8	14.6	14.2	13.4	13.9
	n_{fd}	1.0	0.0	1.0	1.0	2.0	1.0	1.0
$\mathrm{Oct}_{(\mathrm{spr})}$	t_{min}	6.6	6.8	6.3	6.7	5.0	5.9	6.9
	t_{max}	16.8	16.0	15.8	16.8	16.5	15.6	16.3
	n_{fd}	0.0	0.0	0.0	0.0	0.0	0.0	0.0
$\mathrm{Nov}_{(\mathrm{spr})}$	t_{min}	8.5	8.6	8.1	8.3	6.9	7.3	8.7
	t_{max}	18.1	18.1	17.6	18.4	18.5	17.0	18.4
	n_{fd}	0.0	0.0	0.0	0.0	0.0	0.0	0.0
$\mathrm{Dec}_{(\mathrm{sum})}$	t_{min}	9.9	9.9	9.6	9.7	8.3	8.8	10.1
	t_{max}	19.5	19.9	19.4	20.0	20.3	18.5	20.3
	n_{fd}	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Annual	t_{min}	7.3	7.7	7.0	7.1	5.4	6.4	7.7
	t_{max}	16.9	16.8	16.5	16.8	16.7	15.5	17.1
	n_{fd}	9.0	6.0	10.0	8.0	28.0	9.0	11.0

Statistics derived via cell averaging of climatological gridded datasets (2.5 km resolution) sourced from Bureau of Meteorology (2017). Statistics include: t_{min}, daily minimum temperature; t_{max}, daily maximum temperature; n_{fd} number of frost days <0°C. Values in parenthesis denote the seasons: sum, summer; aut, autumn; win, winter; spr, spring.

28 occur for the North East, Coal River Valley, East Coast, Huon/Channel, North West, Tamar Valley and Derwent Valley regions, respectively (Table 1). The Derwent Valley is the coldest wine region with growing season monthly minimums lower by as much as 2°C compared to that of the warmer regions of the Tamar Valley and North East (Table 1). Radiation frosts are the most common type to occur in these areas which often transpire during weather patterns that comprise a cold cloudy day followed by a clear night with little to no wind (Wilson 2001, Halley et al. 2003). They tend to be associated with slow-moving anticyclones with freezing temperature mostly encountered in hilly-undulating terrain and in broad inland valleys that are susceptible to cold air drainage from adjacent slopes (Tait and Zheng 2003, Webb et al. 2016).

Climate data

Daily minimum temperature data of varying temporal resolutions were considered for the analysis. First, long-term weather station data of daily minimum temperature were downloaded from the Australian BoM website portal (http://www.bom.gov.au/climate/data) and appraised for use. Ninety-three stations were analysed and temperature data prior to 2014 were considered (i.e. the majority of the analysis occurred in late 2014), with the aim of using data for the period 1984–2013 (30 years). Ideally, it is preferable to provide 30-year climatic ranges, which are generally accepted for determining long-term climate trends (Déqué 2007); however, due to the irregularity of existing long-term BoM recordings prior to 1994, it was not possible to produce accurate long-term climate data that best represented temperature in the prior decade. Therefore, datasets spanning 20 years (1994 to 2013) were considered for the analysis that produced a reasonable spread of long-term site data with contiguous daily temperature recordings. The recordings were then screened with the criteria that each station has available daily minimum recordings for the entire 20-year period (1994 to 2013). Stations with minor recording gaps (<1 year of missing recordings) were supplemented with interpolated data obtained from recordings of nearby stations using regression estimation as advocated by Sansom and Tait (2004) and Webb et al. (2016). A minimum requirement was set whereby recordings used for supplement recording gaps be highly correlated ($R^2 > 0.8$) to the station with missing data. Stations that did not meet these criteria were removed from this group and considered for integration with the short-term data recordings (refer below). The screening process resulted in a standalone dataset comprising daily long-term (BoM) minimum temperature recordings from 57 long-term site locations (Figure 1).

The BoM climate stations were too sparse to conduct meaningful climate mapping at the desired scale. To complement the existing long-term BoM recordings, additional recording sites were assimilated into the analysis. Thus, 607 additional short-term data sites with a minimum of 12 months of continuous hourly temperature data were acquired and made available courtesy of the Tasmanian Government Department of Primary Industries, Parks, Water and Environment (DPIPWE). The majority of short term sites represent Tinytag temperature loggers (model no. TGP 4017, Gemini Data Loggers, Chichester, England), which were housed in a Datamate Weather Screen (model no. ACS-5050, Hastings Data Loggers, Port Macquarie, NSW, Australia) and protected within a purpose-built steel cage with dimensions measuring 150 × 30 × 30 cm. They were programed to record air temperature (°C) at 1.2 m above ground level every 30 min and were strategically deployed around the state from June 2011 using a stratified sampling approach based on clustering of topographic variables, as per Webb et al. (2016). The data recordings from each were reviewed and underwent quality control in order to remove erroneous data and/or imputating missing values (due to battery failure). Data that did not meet the minimum requirement of continuous hourly recordings for at least a year were discarded. Bureau of Meteorology stations that were initially rejected from the climate station screening process were also grouped with the logger recordings on the provision that a station had at least a year of continuous daily temperature recordings.

The data acquisition and review process resulted in two dataset types being produced: one dataset comprising of short-term temperature data (BoM station and logger recordings) with at least a year of hourly continuous daily temperature recordings (this resulted in 636 site locations across the state, i.e. 607 logger recording sites plus 29 BoM station sites that were initially rejected from the climate station screening process). Another dataset comprising of daily long-term (BoM) temperature recordings (from 1994 to 2013) resulting in 57 long-term climate site locations. Both data types are displayed in Figure 1.

Calibrating short-term site locations to long-term records

The calibration method outlined by Webb et al. (2016) was employed to calibrate long-term (BoM) daily temperature recordings to the locations of the short-term sites. The method involved forming a series of daily BoM grids, which were generated from the sites possessing long-term daily minimum temperature data (refer Figure 1) via a regression kriging interpolation technique (Odeh et al. 1995). The BoM grids were then used in a linear least-squares regression procedure with sites that possessed at least 1 year of short-term daily minimum temperature recordings. Equations that were generated from the regression analysis can be used outside of the short-term recording period to infer estimates from the BoM grids calibrated onto the short-term site locations.

The regression kriging method, as advocated by Webb et al. (2016), was suitable for producing the BoM grids because the number of long-term (BoM) stations satisfied the requirement of at least 50 sites (57 for this study), which were located at a range of distances from each other to sufficiently model spatial autocorrelation (Webster and Oliver 2007). The covariates used for the spatial modelling included the 3-arc sec digital elevation model (DEM; Gallant et al. 2011), the System for Automated Geoscientific Analyses (SAGA) wetness index (Böhner and Antonić 2009, Grabs et al. 2009), a distance to sea index (derived using the Euclidean Distance Spatial Analyst tool in ESRI 2013 ArcGIS Desktop: Release 10.2: Environmental Systems Research Institute, Redlands, CA, USA) and latitude/longitude grids. These were chosen due to their strong relationship with temperature and their ability to account for temperature lapse rates associated with topography (Jarvis and Stuart 2001b, Webb et al. 2016). An average coefficient of determination R^2 of 0.72 confirmed that the daily regression models (calculated using software R, R Development Core Team 2012), were able to account for a significant proportion of the daily minimum temperature variability. The unaccounted variation of the data was modelled using kriging. Henceforth, simple kriging of the regression residual values was then carried out using the gstat module (Pebesma 2004) and the variograms were automatically fitted with exponential models for each subsequent BoM grid produced. The process resulted in 7573 BoM grids in the format of geotiff raster files comprising daily minimum temperature values for the period 1994 to 2013. Bureau of Meteorology grids that coincided with the recordings of the short-term sites were then grouped separately and spatially intersected with the coordinates of the site short-term locations. Days where the BoM grids and the short-term recordings coincide underwent linear least-squares regression (using daily minimum values calculated for the short-term sites vs the corresponding intersected BoM grid cell values) to produce Equation 1 that described the relationship of the BoM grids to the short-term sites:

$$Y = B_1 X + B_0 \tag{1}$$

where Y is the predicted (calibrated) temperature estimate for the short-term site, X is the independent temperature BoM grid raster cell value (estimates extracted from the geographic location of the short-term site), B_0 represents the intercept and B_1 represents the slope. Equation 1 was then subjected to further analysis with the purpose of deriving a history of daily temperature values calibrated to each short-term site. This occurred specifically for days outside of the short-term recording period where the remaining BoM grid cell estimates, previously partitioned from the analysis, was then used as the X variable in Equation 2:

$$Y_{(i,j)} = B_{1\,(i)} X_{(i,j)} + B_{0(i)} \tag{2}$$

where i and j represent the temperature estimate that correspond to the short-term site (i) and the date (j) outside of the short-term recording period. The end product was a calibrated dataset created for each short-term recording site (636 in total) encompassing 20-year series of daily minimum temperature for the period 1994 to 2013. Note that the calibration procedure was carried out for recordings produced in spring (September to November), which was found to marginally improve accuracy compared to using data across an

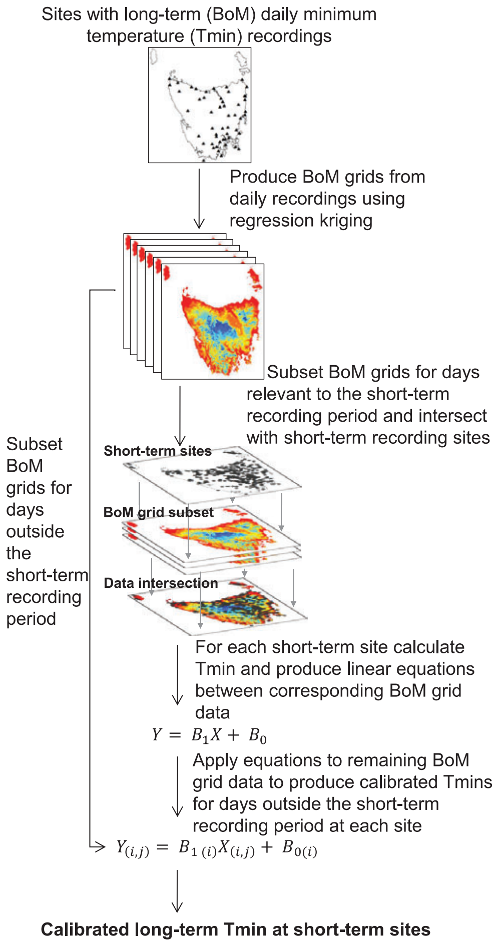

Figure 2. Work flow for calibrating short-term site location to long-term records.

entire year (Webb et al. 2016). Figure 2 outlines the procedure.

Mapping of frost risk

Frost risk during October was determined by counting the frequency years that had at least 1 day of frost occurring at the following minimum temperature thresholds: −3, −2, −1, 0, 1 and 2°C. This was done for all data sites including long-term (BoM) recordings and calibrated short-term sites (as determined above), that is 693 data sites in total. For each site, the frequency of frost that occurred at or below the specified threshold was summed and divided by the total number of years (i.e. 20 years, 1994 to 2013) to obtain the average value and then expressed as a proportion. This is in the form of Equation 3:

$$\mathrm{FR}_{\%} = \frac{\sum_{i=1}^{n} (Fr_i)}{n} \times 100 \tag{3}$$

where $\mathrm{FR}_{\%}$ is the frost risk (%), n equal total number of years relevant to the analysis (i.e. 20 years) and Fr_i:

$$Fr_i = \sum_{i=1}^{31} (T \le T_c) = \begin{cases} 1 \text{ if TRUE,} \\ 0 \text{ else.} \end{cases} \tag{4}$$

represents a binary value (with 1 indicating a frost event) calculated for every year in n (Equation 4), where, i is for each day in October, T the temperature value and T_c represents the temperature threshold. Thus, six spatial point frost risk datasets represented by each of the specified temperature thresholds, T_c, that is 3, −2, −1, 0, 1 and 2°C, were produced.

For each dataset, the Cubist regression tree algorithm (Quinlan 1986) was used to produce continuous frost risk maps. The algorithm is capable of fitting quantitative multivariate relationships between observed properties to covariate data such as those that represent the 'environment' (McBratney et al. 2003). It is analogous to using piecewise linear functions and is an extension of the M5 model tree produced by Quinlan (1992). The process of arriving at a model prediction can be summarised in two stages (Pal and Deswal 2009). The initial stage used a splitting criterion to create a decision tree based on the outcomes of a test that computed the standard deviation of the target values based from a set of training cases C. This was recursive where every potential scenario was evaluated by testing each possible training case as a subset of C, that is C_i. Thus, the aim is to choose a training case that provides the greatest error reduction (ER), which can be summarised as (Equation 5):

$$\mathrm{ER} = \mathrm{SD}(\mathrm{C}) - \sum \frac{|\mathrm{C}|}{|\mathrm{C}|} \mathrm{SD}(\mathrm{C_i}) \tag{5}$$

Where C is a collection of training cases, C_i is a subset of cases that produce an i-th outcome of a test, and SD is the standard deviation. The splitting process often produces large tree-like structures characterised by many subtrees grown on top of parent trees. This can result in overfitting of training values and is mitigated by 'pruning' of the tree. Thus, the second stage of the algorithm involves replacing the subtrees with multivariate linear regression models. These models are based on attributes that were originally referenced by the tests that determined the initial trees and done to ensure the accuracy of the linear model can be objectively assessed. Further simplification of the linear models is then carried out by eliminating parameters that contribute little to the overall model. This reduces multiplicative factors in the partitioning stage (above) and therefore decrease overall estimated error.

For this study, 30 spatial predictor attributes, more formally known as covariates (McBratney et al. 2003, Webb et al. 2016) were used to train the algorithm (i.e. training cases, C) and to guide the mapping across the state. These were projected and resampled to a spatial resolution of 80 m, equivalent to the geographic resolution of the 3-arc sec Shuttle Radar Topography Mission (SRTM) DEM (Gallant et al. 2011). The DEM was used to derive the majority of the covariates using the GIS software SAGA version 2.08 (http://www.saga-gis.org/en/about/references.html) (Conrad et al. 2015). Terrain covariates derived from the DEM included: multi-resolution valley-bottom flatness (MrVBF) (Gallant and Dowling 2003); multi-resolution ridge-top flatness (MrRTF) (Gallant and Dowling 2003); topographic wetness index; SAGA wetness index (SWI); vertical distance to channel network; solar insolation indices (direct/diffuse/duration/total insolation); valley depth; slope; slope height; mid-slope position; sky view factor and visible sky indices; wind effect indexes (windward/leeward effect); analytical hill shade; northness/eastness index (as a surrogate for aspect); effective air flow heights; diurnal anisotropic heating; elevation above channel network; LS-factor; modified catchment area; longitude; latitude; distance to coastline; distance to sea. In addition to these, a persistent greenness index generated from the LandSat 8 satellite imagery was also included and resampled to the equivalent extent and resolution of the DEM.

The spatial point datasets were mapped with the Cubist statistical package (Kuhn et al. 2014). The aforementioned covariates were placed in a raster stack and intersected with the location of the data sites where the accompanying frost risk values calculated for each dataset acted as the training cases, *C*, in the algorithm. To parameterise the predictions of each frost risk Cubist model, the number of trees to grow was set at a maximum of 100 and programed for unbiased rules to be established. As the algorithm uses linear models to predict values within each tree, predictions were allowed to extrapolate 100% outside of the training range. This was in addition to allowing the model to produce up to ten committee models as a way of 'boosting' or generalising models, which was found to be capable of improving predictions by reducing error by up to 5%. For each frost risk Cubist model established, the predictions were then mapped using the covariates to guide the predictions across the entire state. The results are six state-wide geotiff raster datasets representing frost risk at the specified thresholds for October, that is −3, −2, −1, 0, 1 and 2°C.

Validating the frost risk maps with grower surveys

The raster datasets were assessed against expert knowledge via climate survey information (Tasmania State Government 2016) made available to this study by the Tasmanian Government DPIPWE. The survey asked vineyard managers to describe areas of the vineyard where frost was most prevalent as well as the degree of associated risk with regard to damaging frost events during October. Participants were asked to align against pre-defined options for frost event frequency: <1 frost every 20 years (very low risk), 1/20 to 1 frost every 10 years (low risk), >1/10 to 1 frost every 5 years (moderate risk), >1/5 to 1 frost every 2 years (high risk) and 1 > frost every 2 years (very high risk). In proportion terms this corresponded to <5, 5–10, 10–20, 20–50 and >50% of years, respectively. This was in addition to describing the specific dates where crop loss was encountered due to severe frosts. Thirty-two survey responses were received from vineyards located in regions: North West (three responses, vineyard area of 17.5 ha), Tamar Valley (eight responses, vineyard area of 413 ha), North East (seven responses, vineyard area of 68 ha), East Coast (five responses, vineyard area of 270 ha), Coal River Valley (three responses, vineyard area of 114 ha), Derwent Valley (five responses, vineyard area of 79 ha) and Huon/Channel (one response, vineyard area of 3.6 ha). These provided valuable data for ground-truthing the accuracy of the raster outputs.

To align the datasets with the survey responses, individual vineyards were examined spatially. Block extents were manually digitised (using software Google Earth, Version 7.1.2.2041, Google, Mountain View, CA, USA) with particular emphasis placed on zones where frost occurrence was most prevalent (according to detailed descriptions provided by survey participants). These zones were then attributed with frost risk information and converted to a polygon shapefile (using software ESRI 2015 ArcGIS Desktop: Release 10.3) where individual polygons were annotated with the degree of perceived risk as determined by the survey respondents. These data were then intersected with each corresponding frost risk raster dataset where the mean value was calculated as the overall modelled frost risk value for each polygon (for multiple cells that fell within a polygon boundary). Hence, each digitised frost zone comprised of spatial intersects between frost risk survey information and the modelled frost risk output to allow comparisons to be made between the datasets. In order to account for uncertainty between the subjective survey responses and the modelled values, the survey risk ratings for the lower risk classifications were broadened and reclassified. The frequencies of <5% and 5–10% were grouped and reclassified to <10% (low risk) and the classifications of 10–20% and 20–50% grouped and reclassified to 10–50% (moderate to high risk). This served to alleviate some degree of bias that may have been encountered by survey respondents, particularly for the lower risk frequencies, which were comparatively narrow and more prone to underestimation by respondents. The remaining classification of >50% was retained, that is >50% (very high risk). The new grouping was compared to the modelled frost risk values by counting occasions where modelled frost risk values were within a survey classification range as determined by survey respondents. This was expressed as a proportion of agreement value (%) to express the ratio of modelled predictions that aligned with the survey responses.

Results

Assessment of the calibration procedure

To assess the accuracy of the calibration of the short-term sites to the long-term records, a *K*-fold cross validation technique (Hastie et al. 2009) was adopted, as implemented in Webb et al. (2016). This involves separating the training dataset (i.e. days where the BoM grids and the short-term recordings coincide) into *K* equal parts by random sample and using *K* − 1 as the training set and the *K*th part as the validation set. For this study, ten K folds (K = 10) was specified where the following validation metrics were calculated at each *K*-fold and the results were averaged. As a measure of accuracy, the mean absolute error (MAE), was calculated from Equation 6:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|x-y| \tag{6}$$

where x and y represent the true and the predicted value, respectively. The coefficient of determination (R^2) is defined as:

$$R^2 = \left(\frac{n\sum xy-(\sum x)(\sum x)}{\sqrt{\left[n(\sum x^2)-(\sum x)^2\right]\left[n(\sum y^2)-(\sum y)^2\right]}}\right)^2 \tag{7}$$

The concordance coefficient (P_c), defined as:

$$P_c = \frac{2p\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2+(\mu_x-\mu_x)^2} \tag{8}$$

where for μ_x and μ_x represent the means for x and y, respectively, σ_x^2 and σ_y^2 represent the corresponding variances, and p is the correlation coefficient between x and y. Note that P_c is a measure used to quantify the agreement between paired readings that fall on the 45° line through the origin and used to evaluate reproducibility (Lin 1989).

The results of the *K*-fold cross-validation procedure for MAE, R^2 and P_c are shown in Table 2. On average, the MAE

Table 2. *K*-Fold cross-validation averages for calibrating short-term site locations to long-term records.

Statistic	R^2	P_c	MAE (°C)
Mean	0.85	0.91	1.00
SD	0.08	0.05	0.28
Minimum	0.27	0.43	0.39
Maximum	0.96	0.98	2.42

MAE, mean absolute error; P_c, concordance coefficient; R^2, coefficient of determination.

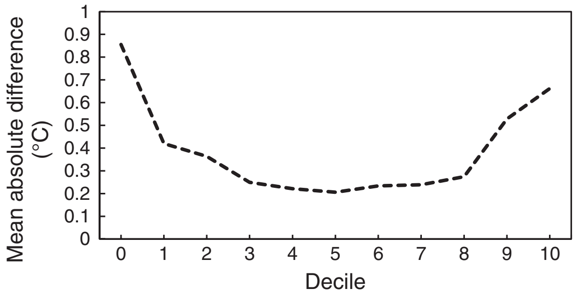

Figure 3. Plot of the mean absolute difference between actual and predicted temperature values for each decile using *K*-fold cross-validation analysis and averaged across all short-term sites.

for calibrating short-term records in spring was 1°C suggesting that errors were consistently within this bound (±1°C). An R^2 of 0.85 and P_c of 0.91 indicated that the predictions were highly correlated with the validation data. As a whole, the validation statistics imply that the calibration method was relatively robust in terms of producing accurate predictions at the short-term recording sites.

Another useful measure to determine the accuracy of the calibration method was to compare the deciles of the actual versus the predicted temperature for all short-term sites. The absolute difference was calculated at each *K*-fold and then averaged to obtain the mean absolute difference (Figure 3). Overall, it was shown that prediction error was within 0.5°C for deciles 1 to 9. This, however, was exceeded by deciles 0 and 10, registering 0.86 and 0.66°C, respectively, suggesting accuracy at the extreme deciles was prone to greater discrepancies.

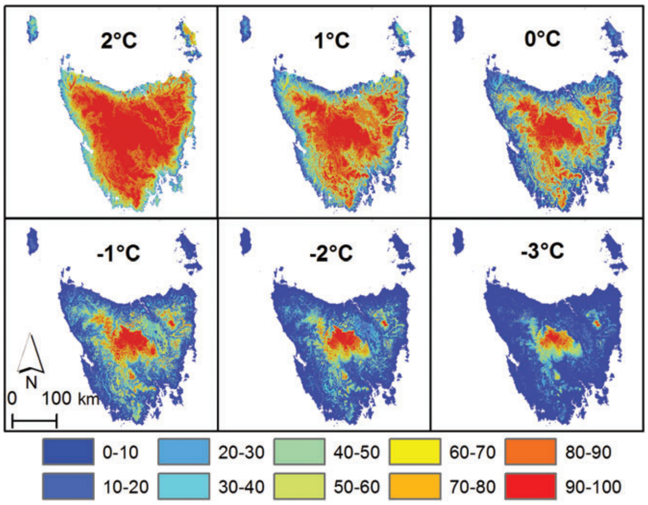

Figure 4. Modelled frost risk (%) outputs for October for minimum temperature thresholds set at –3°C, –2°C, –1°C, 0°C, 1°C and 2°C.

Table 3. *K*-Fold cross-validation averages for modelled frost risk outputs in October at specified minimum temperature thresholds.

Statistic	2°C	1°C	0°C	−1°C	−2°C	−3°C
R^2	0.82	0.84	0.8	0.75	0.76	0.72
P_c	0.9	0.91	0.88	0.86	0.85	0.82
RMSE (%)	12.9	13.8	16.5	16.8	13.2	9.2

P_c, concordance coefficient; R^2, coefficient of determination; RMSE, root mean square error.

Assessment of the frost mapping

Outputs for estimating frost risk across the specified minimum temperature thresholds are displayed in Figure 4. Taken as a whole, the high-risk areas (i.e. 100% modelled risk) generally exist for the inland areas of the state gradually becoming less severe towards the coastal areas. This is especially important for frost risk mapped at −3°C where high-risk areas are more concentrated in centralised/higher elevation areas of the state. For the higher temperature thresholds (i.e. 1 and 2°C), it is clear that frost risk is generally more severe and widespread. This is more pronounced by outputs mapped at the 2°C threshold where the high-risk areas (that is 100% risk) were often predicted within 20 km of the coastline.

To provide insight into the effectiveness of the mapping, as per the calibration procedure, a *K*-fold cross-validation (where K = 10) was implemented to assess mapping accuracy. Each training dataset was separated into *K* equal parts by random sample where *K* − 1 became the new training set and the *K*th part as the validation set. The validation metrics derived at each *K*-fold included the root mean square error (RMSE) (Equation 9):

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(x-y)^2}{n}} \tag{9}$$

where x and y represent the actual frost risk value (as calculated for each data site) and the spatially predicted value, respectively. As similarly calculated for the calibration procedure, the coefficient of determination (R^2) and concordance coefficient (P_c) were also derived. The results of each validation metric were averaged.

The results of the *K*-fold cross validation for RMSE, R^2 and P_c are shown in Table 3. The most accurate predictions in terms of minimising the RMSE were registered by frost risk mapped at −3°C with an RMSE of 9.2%. The least accurate for the month was exhibited by frost risk mapped at −1°C with a comparatively high RMSE of 16.8%. It should be noted that the values for R^2 and P_c were broadly similar across thresholds indicating good agreement between predictions and 'actual' frost risk values despite a relatively high RMSE for outputs such as 0°C.

Comparison of the frost predictions to survey responses

Table 4 presents the results of the modelled frost risk value versus its alignment with the risk classification as determined by the survey respondents for October. Overall, the output for −1°C aligned best with the survey risk classification with 59.4% agreement, followed by the 0°C threshold with 56.3% agreement. The outputs that were least agreeable with the survey responses were 2 and 1°C thresholds with an accuracy of 25.0 and 34.4%, respectively. Output

Table 4. Modelled frost risk values for October at specified minimum temperature thresholds versus survey risk classification as determined by respondents.

VID	Survey risk classification	2°C	1°C	0°C	−1°C	−2°C	−3°C
5	Low (<10%)	20.0	**4.3**	**1.5**	**0.5**	**0.0**	**0.0**
2		26.7	13.6	**1.8**	**0.3**	**0.5**	**0.5**
39		52.1	28.5	**5.6**	**1.1**	**0.7**	**0.4**
7		36.8	18.5	**5.5**	**1.2**	**0.0**	**0.0**
3		46.5	24.3	**8.0**	**3.5**	**1.5**	**2.0**
32		47.0	16.0	**0.3**	**0.0**	**0.0**	**0.0**
9		58.0	37.0	**8.0**	**3.0**	**1.0**	**1.0**
10		62.0	28.3	**2.0**	**1.0**	**0.0**	**0.0**
6		63.0	34.0	11.0	**4.0**	**1.0**	**1.0**
36		64.7	36.0	13.0	**3.0**	**1.0**	**1.5**
34		69.2	37.0	**4.0**	**2.0**	**0.5**	**0.0**
19		70.7	34.9	**4.3**	**2.0**	**0.3**	**0.0**
25		71.0	51.7	28.7	29.0	11.3	**3.0**
17	Moderate–high (10–50%)	**39.0**	**21.0**	5.0	2.0	1.0	1.0
30		**49.0**	**22.0**	6.8	1.0	0.0	0.0
22		51.0	**22.5**	9.5	1.0	0.0	0.0
11		67.0	**45.0**	**16.5**	5.5	1.0	0.0
20		74.5	53.0	**18.5**	9.0	2.5	1.0
8		80.0	57.5	**31.0**	**10.5**	3.0	1.0
31		85.1	53.2	**16.0**	6.4	3.4	0.5
40		63.6	**34.4**	**19.5**	**19.5**	6.1	1.0
12		87.0	65.0	**32.0**	**26.0**	9.0	3.0
18		87.0	70.0	58.0	**27.0**	**12.0**	1.0
14		90.8	69.5	**48.2**	**22.5**	**10.4**	1.8
26		94.5	83.7	**48.3**	**24.0**	7.5	0.0
27		95.5	88.0	50.5	**32.5**	**12.5**	2.5
29	Very high (>50%)	**79.2**	**66.2**	38.0	13.4	5.5	0.7
13		**81.5**	**61.5**	36.5	9.5	3.5	1.0
37		**84.0**	**68.2**	35.9	13.5	5.0	2.5
35		**85.3**	**67.3**	35.8	14.9	5.0	2.0
38		**92.0**	**69.3**	38.0	13.0	5.5	0.7
15		**94.0**	**79.5**	48.5	27.5	12.0	1.5
	Agreement (%)	25.0	37.5	56.3	59.4	46.9	40.6

Values displayed in bold highlight the modelled frost risk values that are within the corresponding surveyed risk classification range corresponding to: low (≤10%); moderate–high (11–50%); very high (>50%). VID, Vineyard Identification Number.

for −3°C tended to agree best with the 'low (<10%)' risk classification with all responses (13 in total) aligning to this classification. This was followed by −1 and −2°C with 12 (for both outputs) and 0°C with ten. For the 'moderate–high (10–50%)' risk classification, 0 and −1°C aligned best at this risk level registering eight and seven, respectively (out of the 13 responses that were encountered for this classification level). In regard to the 'very high (>50%)' risk classification, six out of the six responses aligned to outputs of 2 and 1°C (100% agreement) with none of the other outputs aligning with this classification.

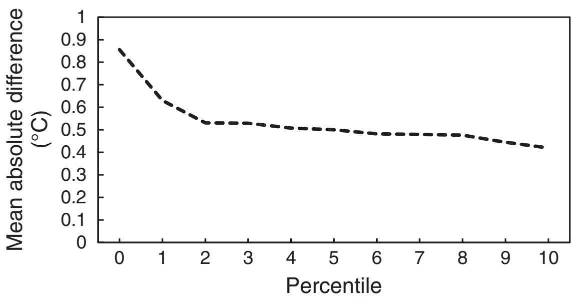

Figure 5. Plot of the mean absolute difference between actual and predicted temperature values for percentiles 0 to 10 using *K*-fold cross-validation analysis and averaged across all short-term sites.

Discussion

Evaluation of the calibration procedure and frost risk mapping

A high degree of accuracy was achieved with the calibration method to infer long-term records onto the short-term sites. From a climate mapping perspective, the MAE difference was within the bounds of what is considered acceptable. Specifically, the mean absolute difference between the actual and modelled temperature was within 0.5°C (Figure 3), as advocated by Sansom and Tait (2004). The only exception to this were high values at the extreme deciles (0 and 10), caused by the larger spread of data at the tail end of the data distributions thereby inflating the standard error of estimation. This was consistent to that encountered in Webb et al. (2016) and Sansom and Tait (2004). However, the relatively high error at decile 0 indicates that extreme low temperature, which is most likely conducive to frost events, could predispose the frost estimates to be inaccurate. A closer look at percentiles 0 to 10 reveals that the inflated error was concentrated at percentile 0 only, which registered 0.86 (Figure 5). Whereas percentiles 1 through to 10 were substantially lower (by a difference of at least 0.2°C in comparison) decreasing steadily from 0.6°C (percentile 1) through to around or just below 0.5°C for the remaining percentiles through to 10. This indicates that extreme minimum temperature was, for the most part, predicted within the vicinity of the accepted mark of 0.5°C, implying that the calibration method for inferring long-term data to each short-term site was relatively accurate for temperature values conducive to frost conditions.

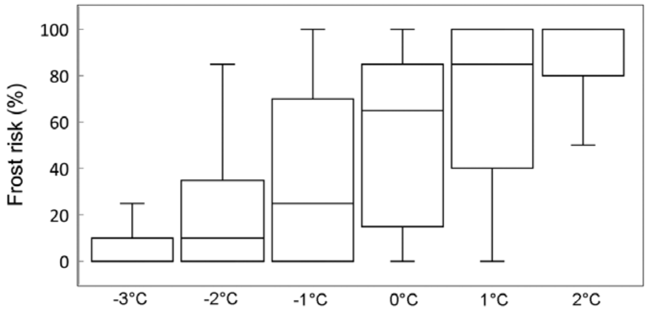

Figure 6. Box and whisker plots showing the distribution of training values (n = 693) for data sites used to interpolate the frost risk outputs.

In regard to the maps of frost risk, the regression tree interpolation method was found to be generally adequate across all temperature thresholds (as displayed in Figure 4). The validation statistics (Table 3) showed that the models were robust in terms of producing good R^2 and P_c values that were similar to that produced by Webb et al. (2016). Of interest, however, was the relatively high RMSE values for some outputs—indicating erroneous predictions—despite the R^2 and Pc measures showing that the predictions were highly correlated with the validation data. As an example, the 0°C threshold displayed an R^2 and P_c value of 0.8 and 0.88, respectively, but a high RMSE of 16.5%. This is contrasted with −3°C, which registered an inferior R^2 and P_c value of 0.72 and 0.82, respectively, but with a lower RMSE of 9.2% (Table 3). In such circumstances, it was expected that a decrease in the RMSE would translate into a proportional upward shift in R^2 and P_c and vice-versa, as demonstrated by Webb et al. (2016). This anomaly was due to a greater proportion of training values skewed to 0 (Figure 6) for −3°C (lower quartile, median and upper quartile values of 0, 0 and 10, respectively) compared to training values for 0°C, which were spread over a wider range (lower quartile, median and upper quartile values of 15, 65 and 85, respectively). In this respect, training datasets representing the lower and upper thresholds (e.g. −3 and 2°C) were based on values, which were largely confined to a narrower prediction interval. This translated to a lower RMSE because the magnitude of prediction errors was minimised by the confined training range.

Nonetheless, the predictions for outputs −1 and 0°C, which exhibited high RMSE but good R^2 and P_c values (Table 3), signify that the regression tree models were generally effective in accounting for frost risk variability. This can be largely attributed to the use of covariate data to form multivariate relationships with the modelled frost risk values. A multiple stepwise linear regression between the frost risk values and the covariate datasets revealed that the two best predictors for nearly all the outputs were the SWI and the DEM, which were similarly utilised heavily in the regression tree algorithm (Table 5). This finding is consistent to that found in Webb et al. (2016), where it was shown that the topographical wetness index (SWI) was a good surrogate for simulating cold air drainage because of its ability to model surface water flowing towards to low point in the landscape, such as valley floors, which is similar to the mechanisms of cold air ponding (Böhner et al. 2002, Chung et al. 2006, Grabs et al. 2009). Frosts of this nature are a regular occurrence for land areas located in southern latitude regions (Tait and Zheng 2003), characterised by still cloudless nights with strong inversions combined with local

Table 5. Important predictors for interpolating frost risk across temperature thresholds as identified by the regression tree algorithm and from stepwise linear regression analysis of the training data versus the covariates.

Temperature threshold	Important covariates	Explained variance (%) from stepwise linear regression analysis
−3°C	SWI (38,50), DEM (79,73)	0.54
−2°C	SWI (39,82), DEM (74,80)	0.58
−1°C	SWI (23,78), DEM (81,86)	0.59
0°C	SWI (22,83), DEM (84,78)	0.61
1°C	DTS (78,77), SWI (12,83)	0.51
2°C	DTS (89,69), SWI (11,67)	0.49

Values in parenthesis denote percent (%) usage by the regression tree algorithm in regards to tree partitioning and model usage, respectively. DEM, digital elevation model; DTS, distance to sea index; SWI, system for automated geoscientific analyses (SAGA) topographical wetness index.

scale advection of dense cold air into low-lying areas (Kalma et al. 1992). In regard to the DEM, it is widely known that elevation strongly influences temperature (Hutchinson 1991, Sansom and Tait 2004, Hunter and Meentemeyer 2005), which explains why the DEM was able to cater for much of the frost risk variability, particularly in mountainous terrain, due to its ability to model the lapse rate with regard to increasing elevation corresponding to decreasing temperature. The DTS index, which was found to be an important predictor for outputs 1 and 2°C, was able to accommodate the influence of the ocean as a natural heat sink which is known to significantly influence minimum temperature for land areas in close proximity to the coast (Trought et al. 1999, Jarvis and Stuart 2001a).

Another way to evaluate the accuracy of the interpolation method was to map minimum temperature values for selected days where frost damage was noted by growers (as garnered from the frost survey) as being particularly severe. This was done using daily minimum temperature from the data sites in conjunction with the regression tree method (as adopted for this study) to interpolate temperature for any day within the 20-year period. Multiple vineyard sites across the state were affected by frost damage on the morning of 16 October 2006, with damage ranging from minor tip burn to complete removal of both primary and secondary buds (Jones et al. 2010). The Derwent Valley site (Figure 7c) especially endured significant damage on the lower areas (80% crop loss) of the vineyard with approximately 30% damage for the higher areas (according to the survey information provided). Figure 7c shows that the interpolation method was able to determine temperature below −2°C for the lower areas and below 0°C for the higher elevation areas, signifying that the mapping was able to delineate the intensity of the event. The Coal River Valley site (Figure 7d) was described as encountering damage with 50% necrosis of shoots and inflorescences (for lower elevation areas) with damage less severe at the top of the slope (Jones et al. 2010). Figure 7d demonstrated that the map was able to determine below freezing conditions for the lower areas of the vineyard justifying why damage for these areas was severe, as detailed in Jones et al. (2010). Accordingly, mapped temperature at the top of the slope was slightly warmer (above 1°C) corresponding to less severe frost damage. The vineyard site portrayed in Figure 7a also exhibited damage with the vineyard encountering up to 70% visible frost damage. In this instance, the entire vineyard was mapped as having temperature

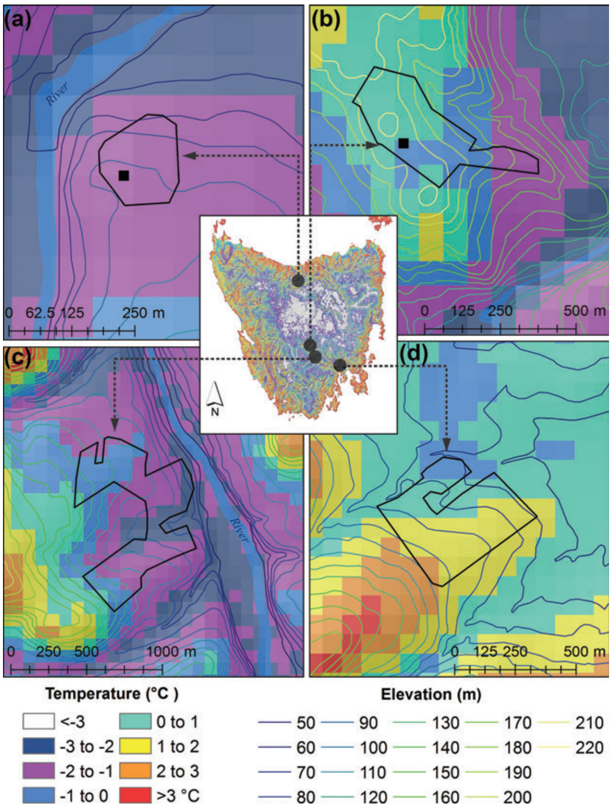

Figure 7. Map showing minimum temperature values on 16/10/2006 as modelled using regression tree interpolation for (a) vineyard site in the North West region, (b) and (c) vineyard sites in the Derwent Valley and (d) vineyard site in the Coal River Valley. Black boundaries denote vineyard extents and the location of temperature recording sites are indicated (■).

values between −1 and −2°C. Records obtained from a weather station located on site (marked by the black square in Figure 7a) showed that the minimum temperature reached −1°C, which is consistent with the mapping. Likewise, for Figure 7b, temperature values garnered on site for this day revealed that minimum temperature reached −0.7°C which is in line with the mapped temperature delineated between 0 and −1°C. Overall, the mapping study was able to account for the severe frost event that occurred on this morning which for many sites constituted an atmospheric inversion of the temperature profile, typical of frost conditions in Tasmania (Perry 1998, Wilson 2001).

Evaluation of the frost predictions to survey responses

The method of aligning the survey responses to mapped frost risk values showed that outputs with thresholds 0 and −1°C were generally favoured. Indeed, a good compromise across all risk categories should imply a temperature threshold that represents frost risk more broadly across the survey responses, and hence, more likely to resemble frost damage as experienced by winegrowers. From this perspective, thresholds of 0 and −1°C were found to correspond well to both low and moderate risk response categories but did not register within the high risk classification. This suggested that there were possible underestimations associated with mapped frost risk values at these locations. This is likely considering that predictions for 0 and −1°C could theoretically be out by as much as 16.45 and 16.81%, respectively (according to the RMSE in Table 3), thereby rendering the predictions to be misaligned by this magnitude. Alternatively, response bias from survey respondents may have also systematically predisposed respondents to favour the most extreme option according to their own personal bias, or paradoxically, influence underestimation to the least severe option (Paulhus 1991). This is compounded further since different cultivars are susceptible to frost damage at various minimum temperature values depending on the time of the frost, environmental conditions and the phenological stage of the crop (Fuller and Telli 1999, Trought et al. 1999, Fennell 2004, Jones et al. 2010, Friend et al. 2011). In other words, respondents may portray the degree of frost damage catered specifically to their own cultivar and management regimes and not reflect frost events consistent with other vineyard sites. Further, the subjectiveness and lack of experience of some respondents may further result in misdiagnosis of frost risk thereby adding to the discrepancies. Nonetheless, the temperature threshold for frost risk mapped at 0 and −1°C is supported by findings of Arias et al. (2010) and Trought et al. (1999) who advocated that grapevines after budburst, particularly at later stages, or after the emergence of new leaves are susceptible to damage between 0 and −1°C. Considering October represents a critical period where similar phenological stages occur for Tasmanian Pinot Noir grapes (Jones et al. 2010), that is the dominant cultivar in Tasmania (Wine Tasmania 2015), it is likely that the 0 and −1°C thresholds broadly align with its susceptibility to frost damage.

To achieve a better measure of the appropriateness of the thresholds, the mapping for the 0 and −1°C outputs (as depicted in Figure 4) was tested visually against the spatial arrangement of digitised frost zones as described by respondents in the survey. The modelled frost risk outputs were spatially classified into their risk classification (i.e. low <10%, moderate–high 10–50% and very high >50%) and then viewed at two discrete locations for vineyards located in the southern zone of the Tamar Valley wine region (Figure 8a,c), and the Cranbrook area of the East Coast wine region (Figure 8b,d). These locations were considered as ideal test locations because they provided a good scenario

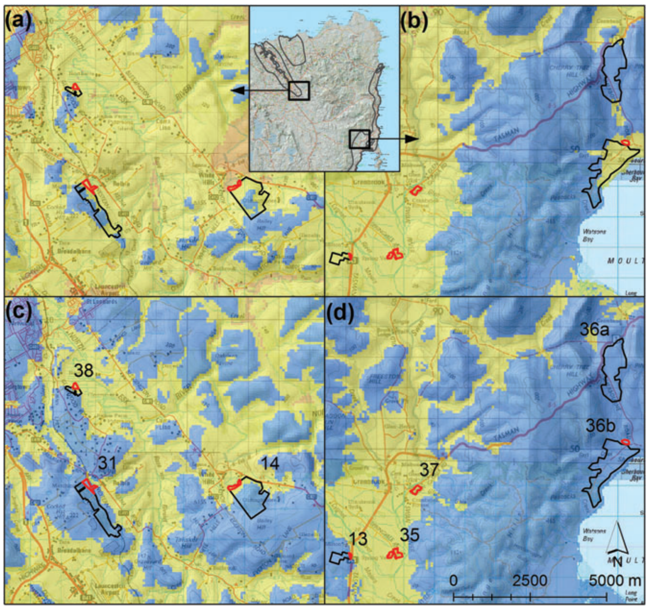

Figure 8. Modelled frost risk outputs for thresholds (a, b) 0°C and (c, d) −1°C for the (a, c) southern Tamar Valley and (b, d) Cranbrook area of the East Coast wine region classified into their risk classifications of low <10% (■), moderate–high 10–50% (■) and very high >50% (■). (c, d) Black boundaries displayed for each panel denote vineyard extents with (c, d) vineyard identification numbers corresponding to values in Table 4. Red boundaries delineate frost prone areas as described by survey respondents.

where low and high frost risk areas were clearly defined (by respondents) and therefore provide a good test where mapped ratings should align to the detailed descriptions. Furthermore, the Cranbrook area is renowned for being plagued by spring frosts (Wilson 2001) and should be reflected by the maps. It was found that the moderate–high risk classification at −1°C tended to align best with the frost prone boundaries in Figure 8c, where the map was able to delineate the frost zones and also recognise the lower risk areas that were on the higher grounds. This was demonstrated for vineyards 38 and 14 (Figure 8c). Likewise, for Figure 8d, the moderate–high rating for the −1°C threshold tended to align best with the digitised frost zones, particularly for vineyard 13 where the frost zone was delineated as moderate–high but the areas in the higher part of the vineyard delineated as low risk. This was consistent with that described by the respondent of this particular site. As such, the 0°C threshold (Figure 8a,b) tended to overestimate the severity whereby the entire vineyard (vineyard 13) was mapped as moderate–high, whereas such areas should only be concentrated to the digitised frost zone in Figure 8c. Moreover, the mapped moderate–high rating at the 0°C threshold in Figure 8b tended to encroach outside of the digitised frost zones more consistently, despite respondents reiterating that such areas were less prone to frost damage. This was particularly over emphasised for vineyard 36b—rated as 'low risk' by the respondent—further showing that frost risk was over accentuated at 0°C. It should be noted that the very high survey ratings provided by respondents for vineyards 38, 37, 35 and 13 (Table 4) did not align with the mapped classification of moderate (Figure 8c,d). As discussed previously, this could be due to uncertainty factors associated with spatial prediction errors, cultivar type or response bias by survey participants. Nevertheless, the moderate risk classification as delineated for both regions at the −1°C threshold align with the observed frost zones within the tested vineyard sites even though anomalies exist between perceived (as determined by respondents) and modelled frost risk.

Application to budburst spatial variability

To further evaluate the appropriateness of the −1°C threshold, frost risk was mapped relative to the average date of budburst across the regions, as similarly conducted by Kartschall et al. (2015) and recommended by Trought et al. (1999). Sufficiently predicting the day of budburst usually requires a database of recorded dates to calibrate and parameterise base temperature and starting time to local conditions (Moncur et al. 1989, Oliveira 1998, García De Cortázar-Atauri et al. 2009, Nendel 2010, Zapata et al. 2017). Limited data exist within Tasmania to satisfactorily calibrate an appropriate model; however, Moncur et al. (1989) provide a useful guide to modelling budburst dates in Australia. Whereby, heat sum units totalling 8200 h above a base temperature of 3.7°C from July was recorded as the requirement to initiate budburst date for Pinot Noir vines in NSW, Australia. This is in the form (Equation 10):

$$\mathrm{HU} = \sum_{h=1}^{24} (Th - Tb) \tag{10}$$

where HU is the heat unit (calculated from July), Th represent the temperature at the hour h, and Tb is the base temperature (3.7°C). By applying the same formula and noting the day of year when 8200 h was reached, B, it was possible

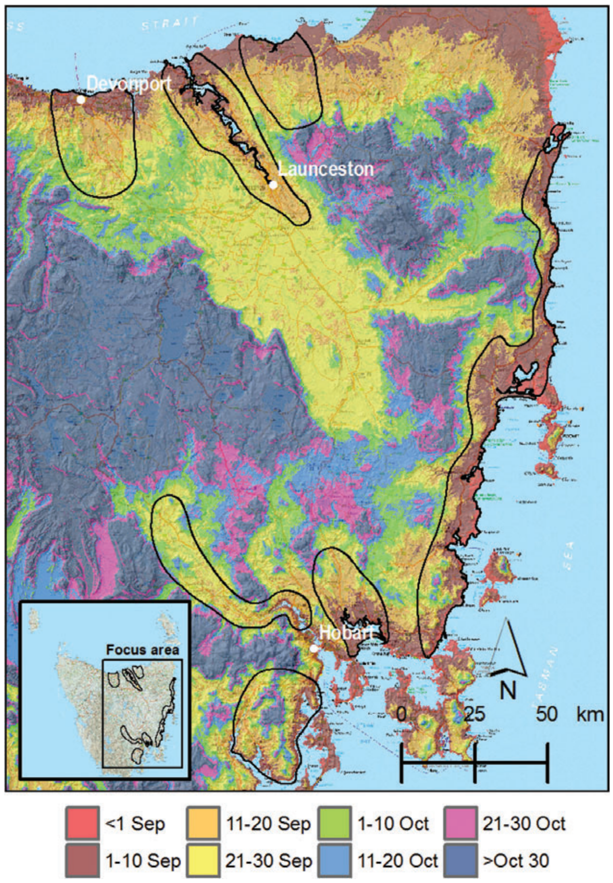

Figure 9. Modelled budburst dates for Pinot Noir vines in Tasmania with recognised wine regions overlaid (black boundaries).

to get an idea of budburst dates across several years at each data site. In this instance, hourly data were used to calculate the heat sums at each data site, derived in a manner similar to that of the minimum temperature but calculated for the 10 years leading into 2013 (due to insufficient long-term BoM hourly recordings prior to 2004). Thus, using Equation 10, budburst dates (day of year) were calculated annually and averaged over the 10-year period to produce average budburst dates at each data site. Interpolation was then carried out using regression trees (as similarly conducted for the frost risk datasets) to produce a continuous map showing average budburst dates depicted in Figure 9.

Validation analysis comparing modelled budburst dates with observed dates for individual years—garnered from phenological observations provided by vineyards for the purpose of this study (Table 6)—provided some indication that predicted dates were in reasonable agreement (with 16 out of the 25 dates predicted within 10 days of recorded budburst). Even though erroneous predictions were made for some years (e.g. 2005/06), the model represented a viable indicator of budburst variability across the regions and was used to reference the first day of the frost vulnerable period in each year. Thus, frost risk (%) relating modelled budburst dates to the date of last significant frost, as similarly advocated by Trought et al. (1999), was calculated for each data site using Equation 3 to derive $FR_{\%}$. Modification to Equation 4 allowed frost events to be calculated on an annual basis, such that Fr_i now indicated when a frost event transpired directly after budburst (in combination with Equation 10) (Equation 11):

Table 6. Budburst dates for Pinot Noir comparing observed and modelled dates for select vineyards across Tasmania's wine regions.

VID	Latitude	Longitude	Region	Observed date *o*	Modelled date *m*	Model error (days) $\lvert o - m \rvert$
7	−42.8	147.44	Coal River Valley	29/08/2005	26/08/2005	3
42	−42.8	147.42	Coal River Valley	18/09/2005	26/08/2005	23
40	−42.75	147.49	Coal River Valley	20/09/2005	29/08/2005	22
				19/09/2006	6/09/2006	13
41	−42.88	147.42	Coal River Valley	13/09/2005	29/08/2005	15
				7/09/2006	4/09/2006	3
15	−42.63	146.85	Derwent Valley	Late September–early October†	23 September‡	NA
18	−42.44	146.74	Derwent Valley	Late September†	24 September‡	NA
13	−42.03	148.05	East Coast	8/09/2013	30/08/2013	9
36	−41.99	148.16	East Coast	Mid-September†	1 September‡	NA
39	−41.07	147.16	North East	22/09/2005	1/09/2005	21
				5/10/2006	8/09/2006	27
4	−41.16	147.23	North East	9/09/2005	15/09/2005	6
				8/09/2013	6/09/2013	2
43	−41.17	147.24	North East	25/09/2010	18/09/2010	7
				3/10/2011	15/09/2011	18
				26/09/2012	20/09/2012	6
				1/10/2013	5/09/2013	26
17	−41.19	147.2	Tamar Valley	Mid-late September†	7 September‡	NA
38	−41.48	146.89	Tamar Valley	22/09/2004	21/09/2004	1
				12/09/2005	4/09/2005	8
				18/09/2006	17/09/2006	1
				15/09/2007	14/09/2007	1
				29/09/2008	27/09/2008	2
				8/09/2009	9/09/2009	1
				22/09/2010	21/09/2010	1
				21/09/2011	10/09/2011	11
				10/09/2012	13/09/2012	3
				2/09/2013	2/09/2013	0

†Indicative period for budburst according to grower observations. ‡Estimate extracted from Figure 9. NA, model not assessed against indicative budburst dates; VID, Vineyard Identification Number.

$$Fr_i = \sum_{i=B}^{D} (if\, T \le T_c) = \begin{cases} 1 \text{if TRUE,} \\ 0 \text{ else.} \end{cases} \quad (11)$$

where B is the modelled budburst date (refer to Equation 10) and, D is the last day of spring, 30 November, which qualified as the end of the frost vulnerable period leading into summer for the wine regions (as indicated in Table 1). Note that the hourly data for the 10 years leading into 2013 were used to calculate frost risk in this manner so consistency was maintained in determining frost events from the date of budburst, B, as similarly conducted by Trought et al. (1999). Regression trees interpolation was used to predict $FR_{\%}$ (as similarly conducted for the October frost risk datasets) to produce continuous map outputs shown in Figure 10.

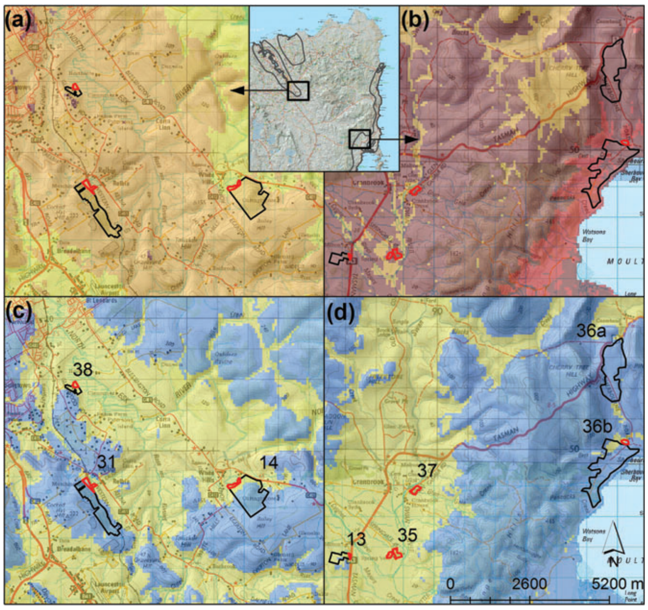

Figure 10. Modelled frost risk outputs (c, d) for the period after budburst for threshold −1°C classified into their risk classifications (refer Figure 8); and modelled budburst dates (a, b) for Pinot Noir vines from Figure 9 for (a, c) the southern Tamar Valley and (b, d) Cranbrook area of the East Coast wine region.

Figure 10c showed that frost risk calculated directly after budburst in the Tamar Valley was similar to that shown for frost risk calculated for October (Figure 8c) and aligned to the frost risk boundaries noted by the survey respondents. Likewise, Figure 10d was also comparable to Figure 8d, with both depicting moderate frost risk severity for vineyards within the moderate frost risk zone in the Cranbrook region. However, it should be noted that frost risk depicted in Figure 10d was modelled as more severe and wide spread compared to that in Figure 8d. This is likely attributed to budburst occurring in early September for the east coast region (as shown in Figure 10b), which would predispose the area to be more susceptible to frost during early spring when temperature is predominately cooler.

With regard to the alignment with the survey risk classifications by survey respondents, $FR_{\%}$ was similar to the −1°C threshold for October (i.e. 59.4%, Table 4), registering the highest value among the tested temperature thresholds with 56.3% agreement (Table 7). The similarity in agreement values indicated that consistency was maintained with the qualitative data across most surveyed vineyard sites. In light of this and in conjunction with the alignment to the frost boundaries depicted in Figure 10c,d, the −1°C threshold was most appropriate with regard to mapping frost risk immediately after budburst. However, it should be

Table 7. Modelled frost risk values for the period after budburst at specified minimum temperature thresholds versus 'survey risk classification' as determined by respondents.

VID	Survey risk classification	2°C	1°C	0°C	−1°C	−2°C	−3°C
5	Low (<10%)	**3.7**	**0.3**	**0.3**	**0.8**	**0.1**	**0.0**
2		66.2	29.6	**9.4**	**0.8**	**0.0**	**0.0**
39		60.0	33.0	**9.6**	**1.0**	**0.0**	**0.0**
7		47.3	26.2	6.0	**4.6**	**0.1**	**0.4**
3		76.5	44.0	20.9	**4.5**	**0.1**	**0.4**
32		58.2	21.3	**6.0**	**0.7**	**0.0**	**0.0**
9		81.2	36.2	16.9	13.6	**1.4**	**0.2**
10		66.1	40.0	18.3	**2.2**	**0.0**	**0.0**
6		54.7	34.6	16.7	12.2	**9.6**	**1.8**
36		73.7	38.6	20.4	10.9	**7.5**	**4.2**
34		72.6	42.9	20.7	**1.6**	**0.0**	**0.0**
19		70.3	33.0	18.6	**3.1**	**0.0**	**0.0**
25		70.6	40.0	17.2	**5.8**	**0.0**	**0.0**
17	Moderate–high (10–50%)	77.9	**40.1**	**17.4**	2.0	0.0	0.0
30		51.1	**16.0**	1.7	0.0	0.0	0.0
22		74.0	**41.7**	**17.7**	6.9	0.8	0.0
11		67.3	**43.5**	**14.3**	1.5	0.0	0.0
20		78.7	52.6	**22.1**	3.7	0.0	0.0
8		88.8	72.4	**31.5**	**13.2**	6.1	1.8
31		88.1	68.5	**35.5**	**11.8**	0.0	0.0
40		77.7	**42.8**	**31.6**	**12.1**	0.2	0.4
12		82.4	53.5	**29.0**	**19.6**	2.5	0.9
18		106.1	85.4	51.6	**25.7**	1.4	0.0
14		96.2	81.6	50.1	**25.6**	4.7	0.2
26		92.7	68.5	**36.1**	**22.1**	4.5	0.0
27		94.4	74.5	**41.6**	**26.2**	**16.0**	5.1
29	Very high (>50%)	**94.7**	**65.8**	37.3	22.2	7.7	0.8
13		**83.6**	**62.1**	29.1	18.3	9.9	0.9
37		**84.4**	**63.2**	31.3	17.3	13.7	4.0
35		**85.4**	**62.9**	31.0	16.5	13.1	6.7
38		**96.9**	**87.2**	**58.2**	32.7	9.2	2.1
15		**100.8**	**89.2**	**53.6**	29.4	13.9	0.1
	Agreement (%)	21.9	37.5	50.0	56.3	43.8	40.6

Values displayed in bold highlight the modelled frost risk values that are within the corresponding surveyed risk classification range corresponding to: low (≤10%); moderate–high (11–50%); very high (>50%). VID, Vineyard Identification Number.

considered that the 10-year period used to model frost risk after budburst may portray variation that is atypical of frost risk over a longer time period and would therefore require longer term data to improve model certainty.

Conclusions

High resolution frost risk maps provide key information for identifying land areas prone to damaging spring and late season frosts. Until now, there has been no consistent method for delineating such areas with a high level of detail and accuracy catered to viticulture suitability. Nor has there been any studies linking frost damage catered to the level of perceived crop loss and risk to winegrowers. The method for calibrating the long-term data onto the short-term data sites as well as the method used for interpolating the data sites was able to demonstrate a high degree of reliability. The interpolation method was further tested for vineyards that had endured historic frost events and was able to replicate temperature below freezing that had predisposed winegrapes to be severely damaged during an event in October 2006. Based on these analyses, the maps were capable of accurately delineating frost risk at the mesoclimate level and ideal for determining viticulture suitability with respect to frost risk.

With regard to aligning the frost risk outputs that best agreed with risk (as provided by survey respondents), it was suggestive that frosts at or below the 0 and −1°C broadly aligned with the experiences and knowledge of growers. The result was inconclusive since multiple factors add a degree of uncertainty including misalignment of risk ratings due to spatial prediction errors, response bias by participants, cultivar types (various resistances to frost) as well as the phenological stage of the vine. Nevertheless, a visual analysis showed that the −1°C threshold and associated risk categories tended to agree with the frost zone extents as described by respondents that had historically been susceptible to frosts events in October (spring). Furthermore, when spatially linked to the day of budburst over a 10-year period, susceptible frost areas as noted by respondents were maintained signifying that the threshold was appropriate in quantifying frost risk when catered to modelled budburst dates (for Pinot Noir vines). It should be noted that eight out of the 11 survey respondents that were located in the moderate–high risk areas (as mapped for the −1°C threshold for Figure 10b,d) reported as having some form of active frost prevention installed. Conversely, vineyards located in the Low-risk areas did not describe adopting any intensive frost prevention measures apart from promoting good air flow movement by ensuring the cover crop was closely mown. Considering there was a reasonable contrast for vineyards located in the moderate–high risk areas with active frost prevention, versus vineyards located in low risk areas without frost prevention, it further justifies that risk classifications for the −1°C threshold were appropriate for this study.

Classifications of <1 frost every 10 years (<10%), >1/10 to 1 frost every 2 years (10–50%) and 1 > frost every 2 years (>50%) correspond well to low, moderate–high and very high risk areas. As frost risk was developed with viticulture suitability in mind, risk ratings with respect to levels of suitability can be linked to suitable, moderately suitable and unsuitable classes corresponding to <1 frost every 10 years (<10%), >1/10 to 1 frost every 2 years (10–50%) and >1 frost every 2 years (>50%). In this regard, an overall suitability map can be achieved by delineating the new suitability classes. Because budburst was found to vary spatially across the wine regions (Figure 9), maps catering to a range of likely budburst dates can be produced using longer term climate data (20 years based on daily minimum temperature). Using Table 6 as a guide, maps catered to early (1 September), middle (16 September) and late (1 October) budburst dates can inform a range of frost risk scenarios linked to suitability for Pinot Noir (Figure 11). For land areas located in the moderately suitable class, it is recommended that frost prevention measures be considered in light of the majority of existing vineyard sites already utilising active mitigation measures in these areas. Adopting measures such as those recommended by Poling (2008) and Trought et al. (1999), including purpose-built sprinkler systems, heaters or wind machines, would ultimately result in such areas becoming more economically viable on the provision that adequate heat accumulation and soil properties are within acceptable ranges as detailed by Kurtural et al. (2007), Hall and Jones (2010) and Yau et al. (2014). Unsuitable mapped areas would require further effort and expense and may be considered economically unviable due to persistent frost events. None of the survey vineyards was located in the unsuitable mapped areas for maps produced in Figure 8b,d and in Figure 10b,d, thus, reinforcing this conclusion. It should be stressed that the suitability and associated risk classifications presented here with respect to the −1°C threshold are indicative and more conclusive studies that quantitatively link actual frost events to damage endured by winegrapes need to be properly distinguished. This should occur for different cultivars at varying temperature thresholds and consistent across phenological stages and management regimes. Improved parameterisation of an appropriate budburst model spatially calibrated to Tasmanian conditions and appropriately linked to frost risk over a longer climatic period would improve mapping certainty.

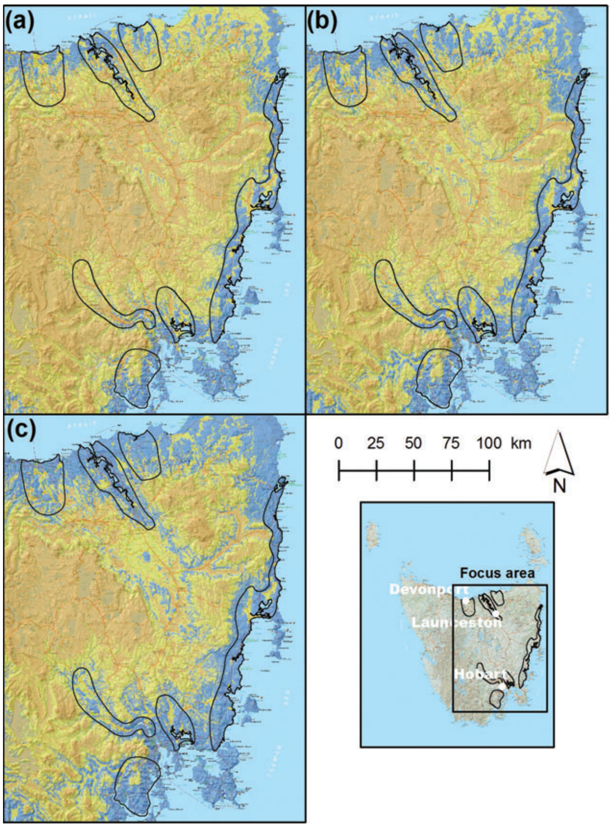

Figure 11. Frost risk suitability for Pinot Noir vines after budburst for dates: (a) 1 September; (b) 16 September and (c) 1 October, delineating suitable (■), moderately suitable (■) and unsuitable (■) classes for land areas in Tasmania's with recognised wine regions overlaid (black boundaries).

Acknowledgements

This research has been supported by the Tasmanian Department of Primary Industries, Parks, Water and Environment (DPIPWE) who have allowed access to temperature logger recordings, frost risk survey information and computing infrastructure to conduct this research. This research was also supported by Tasmanian Partnership for Advanced Computing and by use of the Nectar Research Cloud. The Nectar Research Cloud is a collaborative Australian research platform supported by the National Collaborative Research Infrastructure Strategy (NCRIS). We would like to thank Tasmanian winegrape growers and vineyard managers for providing invaluable assistance and knowledge in assessing our mapping outputs. Special thanks to Dr Fiona Kerslake of the Tasmanian Institute of Agriculture for her contributions in sourcing phenology data and providing valuable insights to the study.

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Manuscript received: 3 March 2017

Revised manuscript received: 23 August 2017

Accepted: 31 August 2017